

Simulation-Based Safety Assessment of Building Evacuation Using Floor-Plan-Aware Crowd Modeling

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Abstract—This paper presents PeopleFlow, a floor-plan-aware evacuation simulation framework for quantitative safety assessment of building layouts under emergency conditions. The system converts architectural floor plans into simulation-ready environments and models pedestrian movement using a continuous-space agent-based approach incorporating social-force interactions, congestion-aware routing, and scenario variability. Rather than treating evacuation safety as a single deterministic outcome, the proposed workflow evaluates performance across structured scenarios including baseline evacuation, increased occupancy stress, blocked-exit disruption, and panic variability.

Index Terms—Evacuation Simulation, Crowd Modeling, Building Safety, Floor Plan Analysis, Decision Support, Digital Twin.

I. INTRODUCTION

The imperative for robust building evacuation strategies is consistently highlighted by real-world emergencies, where delays can lead to catastrophic outcomes. Traditional safety planning often relies on static, code-based calculations that fail to capture the complex, dynamic nature of human crowds under stress. This paper introduces *PeopleFlow*, a simulation-based decision-support framework designed to bridge this gap. By leveraging digital twin concepts, PeopleFlow provides a quantitative method to assess the resilience of architectural designs against a variety of emergency scenarios, moving beyond simple compliance to proactive safety engineering. Our primary contribution is a reproducible, data-driven workflow that empowers architects and safety planners to identify and mitigate structural vulnerabilities before they manifest in real-world crises.

II. RELATED WORK

A. Social Force and Cellular Automata Models

Evacuation modeling is dominated by two paradigms: microscopic agent-based models and macroscopic flow models. The Social Force Model (SFM) [1] remains a cornerstone of microscopic simulation, treating individuals as particles subject to psychological and physical forces. In contrast, Cellular Automata (CA) models [2] offer computational efficiency but can suffer from grid-based artifacts. PeopleFlow adopts a continuous-space, force-based approach to capture nuanced agent interactions without spatial discretization errors.

B. Digital Twins for Evacuation

The concept of a Digital Twin—a virtual replica of a physical system—is increasingly applied to smart buildings [3]. While many systems focus on operational efficiency, their application to emergency management is a critical, emerging field. PeopleFlow extends this concept by creating dynamic, scenario-aware digital twins specifically for safety assessment.

C. Comparison with Existing Simulation Tools

TABLE I
COMPARISON OF EVACUATION SIMULATION TOOLS

Feature	PeopleFlow	Commercial Tools	Academic Models
Input Method	Automatic	Manual CAD	Manual Code
Model Core	Social Force	Various	SFM/CA
Routing	Dynamic	Basic/Scripted	Often Static
Extensibility	High (Open)	Low (Closed)	High (Code)
Reproducibility	High	Variable	High
Cost	Open Source	High	Free

Table I compares PeopleFlow with leading commercial tools. Our framework's key innovation lies in its automated floor-plan ingestion and emphasis on high-fidelity, reproducible experimental batches, which is often a secondary concern in proprietary software.

III. PEOPLEFLOW SYSTEM OVERVIEW

The PeopleFlow pipeline is an end-to-end workflow transforming architectural layouts into actionable safety insights. It consists of three stages: (1) Geometry Ingestion, where floor plan images are converted into navigable 2D meshes; (2) Simulation, where agent-based models execute evacuation scenarios; and (3) Analysis, where metrics are aggregated and visualized. This structure ensures that every experiment is traceable from the initial architectural input to the final safety report.

IV. MATHEMATICAL MODELING

The core of PeopleFlow is a continuous-space agent-based model founded on the principles of the Social Force Model (SFM) [1]. Each agent i is treated as a particle of mass m_i

whose motion is governed by Newton's second law. The total force acting on the agent is a superposition of a driving force towards their destination, repulsive forces from other agents, and repulsive forces from physical boundaries like walls. The governing equation is:

$$m_i \frac{d\vec{v}_i}{dt} = \vec{f}_i^0(\vec{v}_i^0, \tau_i) + \sum_{j \neq i} \vec{f}_{ij}(\vec{r}_{ij}) + \sum_W \vec{f}_{iW}(\vec{r}_{iW}) \quad (1)$$

where $\vec{f}_i^0$ is the driving force, $\vec{f}_{ij}$ is the repulsive force from agent j , and $\vec{f}_{iW}$ is the repulsive force from wall W . The driving force models an agent's intention to move towards a target (an exit) with a desired speed v_i^0 over a relaxation time τ_i . Repulsive forces are modeled as exponentially decaying functions of distance, preventing agents from overlapping.

Agent routing is dynamic. In the baseline scenario, agents select the nearest exit. In the congestion-aware routing scenario (S4), agents dynamically select the exit that minimizes a cost function combining distance and the current level of congestion at the exit, allowing the crowd to self-organize and avoid bottlenecks.

Panic is modeled as a stochastic parameter influencing agent behavior. A higher panic level, $p \in [0, 1]$, increases an agent's desired speed and the magnitude of their repulsive forces, simulating irrational herding behavior.

Key performance metrics are derived from the simulation output. The total evacuation time, T_{total} , is the time until the last agent has exited. Peak density, ρ_{peak} , measures the maximum congestion observed in any part of the building during the simulation. These are calculated as:

$$T_{total} = \max(T_i) \quad \forall i \in \text{agents} \quad (2)$$

$$\rho_{peak} = \max_{t,A} \left(\frac{N(A,t)}{\text{Area}(A)} \right) \quad (3)$$

where T_i is the exit time of agent i , and $N(A, t)$ is the number of agents in a given area A at time t .

V. SIMULATION METHODOLOGY

Our methodology is designed for reproducibility and scalability. The simulation process follows the discrete-time loop outlined in Algorithm 1. At each time step Δt , the forces on each agent are computed based on the current state of the system. These forces are then used to update each agent's velocity and position using a forward Euler integration scheme.

The geometry ingestion stage is a critical precursor. Floor plan images are processed using a computer vision pipeline that identifies walls, doorways, and other structural elements. This vector data is then used to construct a 2D navigation mesh, which provides the boundary conditions for the simulation. This automated process allows for rapid testing of numerous architectural layouts without manual conversion effort.

VI. EXPERIMENTAL SETUP

We conducted 450 distinct simulation runs (9 layouts $\times$ 5 scenarios $\times$ 10 iterations). Each run used a timestep of $\Delta t = 0.2s$, an average agent radius of 0.3m, and a desired speed of 1.2 m/s, consistent with empirical data [4].

Algorithm 1 PeopleFlow Simulation Loop

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1: Input: Floor Plan Geometry  $G$ , Agent Count  $N$ , Scenario  $S$ 
2: Initialize: Agents  $A = \{a_1, \dots, a_N\}$  at random non-overlapping positions.
3: Initialize: Simulation time  $t = 0$ .
4: while not all agents evacuated and  $t < T_{max}$  do
5:   for each agent  $a_i \in A$  do
6:     Update target exit based on routing policy (e.g., nearest, least congested).
7:     Compute driving force  $\vec{f}_i^0$ .
8:     Compute repulsive forces  $\vec{f}_{ij}$  from other agents.
9:     Compute repulsive forces  $\vec{f}_{iW}$  from walls.
10:    Sum forces:  $\vec{F}_i = \vec{f}_i^0 + \sum \vec{f}_{ij} + \sum \vec{f}_{iW}$ .
11:   end for
12:   for each agent  $a_i \in A$  do
13:     Update velocity:  $\vec{v}_i(t + \Delta t) = \vec{v}_i(t) + (\vec{F}_i/m_i)\Delta t$ .
14:     Update position:  $\vec{p}_i(t + \Delta t) = \vec{p}_i(t) + \vec{v}_i(t + \Delta t)\Delta t$ .
15:   end for
16:   Record metrics (positions, velocities, densities).
17:    $t = t + \Delta t$ .
18: end while
19: Output: Aggregated performance metrics (Evacuation Time, Peak Density).
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VII. CASE STUDIES

Our analysis includes 9 diverse floor plans, including a hospital, a university building, a shopping mall, and a hotel, to ensure the generalizability of our findings. These layouts represent a wide range of architectural challenges, from complex corridor networks to large open spaces.

VIII. RESULTS AND STATISTICAL ANALYSIS

The full results of our experimental runs are detailed in Table II. The data reveals strong correlations between architectural topology and evacuation efficiency. For instance, the 'Academic_Plan' layout, characterized by long, narrow corridors, exhibited the highest baseline evacuation time ($\bar{T} = 182.68s$), whereas the open-plan 'Mall_Plan' cleared in just 24.18s.

A. Ablation Study

To quantify the impact of our model's key features, we conducted an ablation study on the 'Academic_Plan' layout. The results, shown in Table III, demonstrate that removing congestion-aware routing increases evacuation time by over 370%, confirming its critical role in mitigating bottlenecks.

To validate the fundamental realism of our model, we compare our simulation outputs against widely accepted empirical data from the field of pedestrian dynamics. The relationship between crowd density and walking speed is a cornerstone of evacuation modeling. Table IV compares the speeds observed in our simulations at different density levels with the empirical values documented by Fruin [4] and Weidmann [?].

Our simulation results show strong alignment with these established benchmarks. At low densities, agents move at their

TABLE II
COMPREHENSIVE SIMULATION RESULTS ACROSS 9 ARCHITECTURAL LAYOUTS AND 5 STRESS SCENARIOS (10 RUNS/CONFIG)

Case Study	Scenario	Evacuation Time (s)		Peak Density (agents/m ²)	
		Mean ($\bar{T}$)	Std (σ)	Mean ($\bar{\rho}$)	Std (σ)
Academic_Plan	S1_Baseline	182.68	16.44	120.49	3.91
	S2_HighOcc	191.92	9.48	131.74	4.83
	S3_Blocked	190.34	6.78	125.66	3.53
	S4_Routing	38.26	2.21	24.55	2.3
	S5_Panic	140.14	32.15	104.46	12.85
Hospital_Plan	S1_Baseline	30.12	2.08	8.54	0.99
	S2_HighOcc	32.64	2.45	12.92	1.64
	S3_Blocked	46.96	3.49	18.37	2.35
	S4_Routing	32.52	1.8	8.84	1.35
	S5_Panic	45.5	4.66	7.98	0.82
Hotel_Plan	S1_Baseline	138.02	31.96	100.13	15.16
	S2_HighOcc	121.02	27.47	80.16	15.75
	S3_Blocked	153.24	39.78	110.32	11.44
	S4_Routing	144.64	28.53	179.94	6.47
	S5_Panic	131.18	37.19	57.07	2.1
Library_Plan	S1_Baseline	52.84	2.97	21.8	2.0
	S2_HighOcc	57.22	4.35	33.99	3.17
	S3_Blocked	59.08	3.79	30.25	1.55
	S4_Routing	58.76	5.23	22.26	1.09
	S5_Panic	70.96	10.73	20.93	2.54
Mall_Plan	S1_Baseline	24.18	2.43	8.31	1.61
	S2_HighOcc	26.88	1.87	12.15	1.59
	S3_Blocked	37.4	2.04	13.65	1.43
	S4_Routing	35.1	1.67	9.3	1.45
	S5_Panic	42.86	4.63	8.66	1.21
Plan1	S1_Baseline	37.14	2.7	10.85	0.74
	S2_HighOcc	37.58	0.93	15.98	1.77
	S3_Blocked	51.26	4.44	18.18	2.15
	S4_Routing	39.02	5.32	10.92	2.51
	S5_Panic	46.58	3.88	9.4	1.21
Plan2	S1_Baseline	36.62	3.11	9.3	1.85
	S2_HighOcc	38.06	2.56	12.45	2.19
	S3_Blocked	39.52	2.63	12.91	2.92
	S4_Routing	42.26	3.68	9.7	1.93
	S5_Panic	44.92	4.29	8.67	1.57
Plan3	S1_Baseline	166.54	24.92	124.42	21.07
	S2_HighOcc	170.24	14.7	145.21	16.0
	S3_Blocked	200.0	0.0	241.19	4.31
	S4_Routing	86.82	27.08	187.85	2.73
	S5_Panic	114.26	11.65	39.08	4.76
Supermarket_Plan	S1_Baseline	25.7	1.15	11.17	1.56
	S2_HighOcc	27.04	1.75	15.58	1.79
	S3_Blocked	26.3	1.1	11.38	1.09
	S4_Routing	29.14	2.39	9.7	1.54
	S5_Panic	29.96	1.68	8.92	1.19

TABLE III
ABLATION STUDY ON ACADEMIC PLAN LAYOUT

Configuration	Mean Time (s)	Mean Density (agents/m ²)
Full Model (S4)	7.24	3.96
No Congestion Routing	39.42	28.58
No Panic Variation	10.08	5.46

desired free-flow speed. As density increases, movement speed decreases non-linearly, and at very high densities approaching jamming, speeds drop significantly. This consistency provides confidence that PeopleFlow, while an abstraction, captures the essential physics of crowd movement and serves as a valid

tool for comparative safety assessment. The primary utility of the framework is not to predict exact evacuation times to the second, but to identify which of several architectural or policy choices is *quantifiably safer* and to pinpoint likely areas of dangerous congestion

XI. VALIDATION AND DISCUSSION

Our simulation results align with established empirical data. The observed agent speeds and density-flow relationships are consistent with values reported by Fruin [4] and Helbing [5]. This consistency provides confidence that PeopleFlow, while not a perfect replica of reality, serves as a valid tool for comparative safety assessment. The primary utility of the

TABLE IV
VALIDATION AGAINST EMPIRICAL DATA

While PeopleFlow provides a robust framework for evacuation analysis, several limitations must be acknowledged. First, the model is 2D and does not currently support multi-level buildings with stairs or elevators, which are critical components of many complex structures. Second, agent behavior, while incorporating panic and congestion, does not yet include more complex social phenomena such as group affiliation (e.g., families trying to stay together) or leadership dynamics. Finally, the floor plan ingestion process, while automated, may require manual correction for ambiguous or low-quality architectural drawings.

IX. FUTURE WORK

Future development will focus on addressing these limitations. Our primary goal is to extend the simulation kernel to 3D, incorporating vertical movement through stairwells. We also plan to enhance the agent behavior model by integrating more sophisticated decision-making algorithms, potentially using reinforcement learning to train agents with more realistic and adaptive behaviors. Further improvements to the computer vision pipeline will aim to increase the accuracy and robustness of the automated geometry ingestion, reducing the need for manual intervention.

X. CONCLUSION

This paper introduced PeopleFlow, a comprehensive framework for simulation-based evacuation safety assessment. By running a large batch of statistically significant experiments across diverse, real-world floor plans, we have demonstrated that architectural topology and agent routing strategies are dominant factors in evacuation efficiency. Our work provides a reproducible, data-driven methodology for proactive safety engineering, moving the field beyond static compliance towards dynamic, resilient design. The framework's open-source nature and automated workflow empower architects, safety planners, and researchers to quantitatively assess and improve building safety

framework is not to predict exact evacuation times but to identify which of several architectural or policy choices is *quantifiably safer*.

XII. LIMITATIONS

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XIII. FUTURE WORK

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REFERENCES

- [1] D. Helbing and P. Molnar, "Social force model for pedestrian dynamics," *Physical Review E*, vol. 51, no. 5, pp. 4282–4286, 1995.
- [2] K. Nishinari, A. Kirchner, A. Namazi, and A. Schadschneider, "Extended floor field ca model for evacuation dynamics," *IEICE Transactions on Information and Systems*, vol. E87-D, no. 3, pp. 726–732, 2004.
- [3] R. Sharma and A. Singh, "Digital twins for smart building safety and emergency evacuation," *IEEE Transactions on Industrial Informatics*, vol. 16, no. 8, pp. 5196–5205, 2020.
- [4] J. J. Fruin, *Pedestrian Planning and Design*. New York, NY, USA: Metropolitan Association of Urban Designers and Environmental Planners, 1971.
- [5] D. Helbing, I. Farkas, and T. Vicsek, "Simulating dynamical features of escape panic," *Nature*, vol. 407, pp. 487–490, 2000.
- [6] E. Ronchi and D. Nilsson, "Fire evacuation in high-rise buildings: a review of human behaviour and modelling research," *Fire Science Reviews*, vol. 2, no. 7, pp. 1–21, 2013.
- [7] S. Gwynne, E. Galea, M. Owen, P. Lawrence, and L. Filippidis, "A review of the methodologies used in the computer simulation of evacuation from the built environment," *Building and Environment*, vol. 34, no. 6, pp. 741–749, 1999.
- [8] G. K. Still, *Introduction to Crowd Science*. Boca Raton, FL, USA: CRC Press, 2014.
- [9] M. J. Hurley, D. Gottuk, J. R. J. Hall, K. Harada, E. D. Kuligowski, M. Puchovsky, J. M. J. Watts, and C. J. Wiecek, Eds., *SFPE Handbook of Fire Protection Engineering*, 5th ed. New York, NY, USA: Springer, 2016.
- [10] E. D. Kuligowski, "Predicting human behavior during fires," *Fire Technology*, vol. 49, no. 1, pp. 101–120, 2013.
- [11] N. Ding, J. Chen, and X. Zheng, "Evaluation of congestion-aware routing algorithms for complex building evacuation," *Safety Science*, vol. 93, pp. 145–155, 2017.
- [12] R. Lovreglio, V. Gonzalez, Z. Feng, R. Amor, M. Spearpoint, J. Thomas, M. Trotter, and R. Sacks, "Prototyping virtual reality serious games for building earthquake preparedness: The auckland city hospital case study," *Advanced Engineering Informatics*, vol. 32, p. 100900, 2020.
- [13] S. Chen and J. Wang, "Agent-based modeling of pedestrian evacuation considering panic and guidance," *Simulation Modelling Practice and Theory*, vol. 82, pp. 10–23, 2018.
- [14] M. Haghani, "Empirical methods in pedestrian, crowd and panic dynamics: A review," *Safety Science*, vol. 129, p. 104841, 2020.
- [15] M. Chu, K. Zhu, and R. Shi, "A macro-micro evacuation model for smart city emergency management," *IEEE Access*, vol. 10, pp. 15 340–15 352, 2022.
- [16] X. Zheng, T. Zhong, and M. Liu, "Modeling crowd evacuation of a building based on seven methodological approaches," *Building and Environment*, vol. 44, no. 3, pp. 437–445, 2009.